

1. Give the element for designing bus.

Element for Designing Bus:

- ✓ Data Bus: Transfers data between the CPU, memory, and peripherals.
- ✓ Address Bus: Specifies the location in memory for data transfer.
- ✓ Control Bus: Manages data flow and communication control signals.

2. Why study COA? Describe the structure and functions of a computer.

- To understand how computers work at the hardware level, optimizing performance and designing efficient software.
- **Structure of a Computer:**
 - ✓ CPU (Central Processing Unit): Executes instructions.
 - ✓ Memory (RAM): Stores data and instructions temporarily.
 - ✓ Input Devices: Take in data (keyboard, mouse, etc.).
 - ✓ Output Devices: Display or output processed information (monitor, printer, etc.).
 - ✓ Storage Devices: Store data permanently (hard drives, SSDs).
- **Functions of a Computer:**
 - ✓ Fetch: Retrieve instructions from memory.
 - ✓ Decode: Interpret the instructions.
 - ✓ Execute: Perform the operation.
 - ✓ Store: Save results back in memory.

3. Describe the evolution of DRAM and processor characteristics.

▪ **DRAM Evolution:**

First Generation DRAM (1970s):
Low density, slow, high power.

Fast Page Mode DRAM (Late 1980s):
Improved speed, page mode access.

Extended Data Out DRAM (Early 1990s):
Faster access times, optimized data output.

Synchronous DRAM (Mid-1990s):
Synced with system clock, higher transfer rates.

Double Data Rate Synchronous DRAM (Late 1990s):
Doubled data transfer, increased bandwidth.

▪ **Processor characteristics:**

Early Processors:
Single-core, low clock speeds, basic instruction sets.

Modern Processors:
Multi-core, high clock speeds, advanced instruction sets, power-efficient.

4. Prepare a question on semiconductor main memory and answer it yourself.

Question: What is the primary function of semiconductor main memory in a computer system?

Answer: Semiconductor main memory, often in the form of RAM (Random Access Memory), serves as a temporary storage location for both data and instructions that the CPU actively uses during operations. Unlike permanent storage (e.g., hard drives), semiconductor main memory allows for quick access and retrieval of information, contributing to the overall speed and efficiency of a computer system. It is volatile, meaning it loses

its contents when power is turned off, but its fast read and write capabilities make it crucial for the smooth functioning of a computer.

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5. ~~Give the basic elements to design different buses.~~ Describe PCI bus structure.

The PCI (Peripheral Component Interconnect) bus structure consists of a shared parallel bus architecture used for communication between various hardware components within a computer system. Key features include:

- Bus Topology: Parallel bus design for simultaneous data transfer.
- Address and Data Lines: Separate address and data lines for efficient communication.
- Bus Width: Initially 32 bits wide, later extended to 64 bits for higher bandwidth.
- Clock and Synchronization: Synchronous operation with a system clock for data synchronization.

6. Mention the characteristics of computer family. **Distinguish among sequential, nested, time sequence and priority interrupts.**

- ✓ **Common Architecture:** Share a fundamental design structure.
- ✓ **Instruction Set Compatibility:** Use similar or compatible instruction sets.
- ✓ **Common Operating System:** Support the same or compatible operating systems.
- ✓ **Unified Peripheral Devices:** Utilize similar peripheral devices and interfaces.
- ✓ **Interchangeable Software:** Run software applications across the family with minimal modification.

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7. Describe peripheral devices.

Peripheral devices are external add-ons to a computer, expanding its functions. They include input (e.g., keyboard), output (e.g., printer), and storage devices (e.g., external hard drives), connecting through interfaces like USB. They allow customization and enhance overall functionality.

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1.

a) What, in general terms, is the distinction between computer organization and computer architecture? List and briefly define the main structural components of a computer.

➤ **Distinction between CO and CA.**

- Computer architecture refers to the attributes visible to the programmer and which affect the logical execution of a program.
- Computer organization refers to the hardware units that realize the architecture: the instruction set, the number of bits used to represent data types, addressing memory, etc.

➤ **Main structural components of a computer:**

- CPU: Controls the operation of the computer and performs its data processing functions.
- Main memory: Stores data.
- I/O: Moves data between the computer and its external environment.
- System interconnection: Some mechanism (e.g. a system bus) that provides for communication among CPU, main memory, and I/O.

b) Explain Moore's law. On the IAS, describe the process that the CPU must undertake to read a value from memory and to write a value to memory in terms of what is put into the MAR, MBR, address bus, data bus and control bus.

- **Moore's Law** states that the number of transistors on a microchip doubles every two years. The law claims that we can expect the speed and capability of our computers to increase every two years because of this, yet we will pay less for them.
- **For read** a value from memory, CPU will put the address of the value it wants into MAR and asserts the read control line to memory in order to place the address on the address bus. Memory will place the contents of memory location passed on the data bus and then transferred to MBR. For write a value to memory, CPU have to put the address of the value it want to write into MAR. CPU will put the data it want to write into MBR Then CPU will run the write control line to memory and places the address on the address bus and data on the data bus. Then only memory will transfers the data on the data bus into the corresponding memory location

c) What are the key properties of semiconductor memory? What is the difference between DRAM and SRAM in terms of characteristics.

- **Key properties of semiconductor memory:**
 - ✓ Stores data in magnetic form.
 - ✓ Affected by magnetic fields.
 - ✓ Has high storage capacity.
 - ✓ Doesn't use a laser to read/write data.
 - ✓ Magnetic storage devices are; Hard disk, Floppy disk, Magnetic tape etc.

➤ **SRAM VS DRAM**

SRAM	DRAM
It can store data as long as electricity is available.	It saves data for as long as the power is on or for a few moments if the power is turned off.
SRAM has a storage capacity of 1 MB to 16 MB in most cases.	DRAM, which is often found in tablets and smartphones, has a capacity of 1 GB to 2 GB.
The storage capacity of SRAM is low.	The storage capacity of DRAM is higher than SRAM.
SRAM is more expensive than DRAM.	DRAM is less expensive than SRAM.
It is comparatively faster.	It is comparatively slower.
SRAM is used in cache memories.	DRAM is used in main memories.

d) Define memory cell, Describe the structure of a typical memory cell.

- **A memory cell** is a small unit of storage in a computer's memory, typically used to store a single byte of data.

Today, the most common memory cell architecture is MOS memory, which consists of metal–oxide–semiconductor (MOS) memory cells. Modern random-access memory (RAM) uses MOS field-effect transistors (MOSFETs) as flip-flops, along with MOS capacitors for certain types of RAM